

RESEARCH PROFILE

Seeking a funded Doctoral Researcher or Research Assistant position in human-computer interaction, human-AI interaction, immersive systems, mixed reality, or user research, with availability from August–September 2027.

Human-computer interaction researcher studying what happens between an AI contribution and a person's next move. Builds instrumented systems and traces how people interpret, reshape, adopt, or reject AI-generated material across proactive AI and AR/XR. Combines controlled experiments, interaction-log analysis, mixed methods, and development to distinguish changes in system output from changes in human thinking, agency, and experience.

RESEARCH AND PROFESSIONAL EXPERIENCE

- AXIS Lab, Interaction Science Institute, Sungkyunkwan University 2024–present
Undergraduate Research Assistant (2024–Feb 2025); Master's Student Researcher (Mar 2025–Aug 2026); Researcher (Sept 2026–present). Contributing to projects on proactive AI and human cognitive mechanisms in always-on AI environments, and continuing prototype-based research on AR smart glasses, socially visible sensing, and immersive human-AI interaction.
- Researcher and App Developer, Personalized AI Story Generation Project Apr 2026 – present
Developing Phodong, a production-stage SwiftUI/Firebase system that turns children's real-world object discoveries into personalised stories. Includes age-adaptive generation, parent-gated controls, and an instrumented discovery-to-story pipeline for a planned family study; supported by Seoul RISE and led by Prof. Changjun Lee.
- Assistant Instructor, Generative AI Content Creation Workshop, Hyundai Futurenet Jul 2026
Supported hands-on workshop activities on generative-AI content creation, including participant guidance and practical exercises for producing AI-assisted visual and interactive media.
- Research Intern, RGB Makers Co., Ltd. Aug 2025
Analysed first-person interaction data and privacy risks in metahuman-based content, developed a MetaHuman character in Unreal Engine 5, and proposed outdoor interaction scenarios for user-centred immersive experiences.
- BK21 Research Assistant, Social Interaction Big Data Education and Research Group Mar – Jun 2025
Supported immersive-device user research at AXIS Lab through literature review, study and protocol preparation, participant recruitment materials, lab setup, research ethics documentation, and data management.

EDUCATION

- Sungkyunkwan University · Master of Science in Engineering, Immersive Media Engineering Mar 2025 – Aug 2026
Advisor: Prof. Changjun Lee · AXIS Lab · Cumulative GPA: 4.45/4.50 (99.5/100)
Thesis: *When Sensing Becomes Socially Visible: Wearer Meaning and Adoption Resistance in AR Smart Glasses*
- Sungkyunkwan University · Bachelor of Culture & Technology; Bachelor of Economics Feb 2021 – Feb 2025
Major in Culture and Technology Convergence; International Trade Linked Major, Economics Track. Cumulative GPA: 4.32/4.50 (98.2/100). Graduated with SKKU Summa Cum Laude.

PUBLICATIONS

PEER-REVIEWED CONFERENCE PUBLICATIONS

- Kim, G., Choi, Y., Kim, Y., & Lee, C. (2026).** No Time to Say Goodbye: Emotional Loss Responses to Sudden Termination in Immersive AI Interactions. *CHI EA '26: Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. ACM. DOI: [10.1145/3772363.3798687](https://doi.org/10.1145/3772363.3798687)
- Kim, G., Choi, Y., & Lee, C. (2025).** See, Sense, Spark: Profiling Granularity in Proactive AI for Creative Research Ideation. *SIGGRAPH Asia 2025 Posters*. ACM. DOI: [10.1145/3757374.3771458](https://doi.org/10.1145/3757374.3771458)
- Kim, G., & Han, Y. (2024).** Camp_Around: An Integrated Tour Service for International Students in Korea. *HCI Korea 2024 Creative Award Proceedings*, 1224–1227.

JOURNAL ARTICLES

- Kim, G., Kim, S. Y., Park, S., Park, S., Choi, Y., & Lee, C. (2025).** What Makes Audiences Pay More for Immersive Cinema? A Conjoint Analysis of ScreenX Preferences and Willingness-to-Pay. *International Telecommunications Policy Review*, 32(4), 1–45. DOI: [10.37793/ITPR.32.4.1](https://doi.org/10.37793/ITPR.32.4.1)
- Kim, G., Choi, Y., & Lee, C. (2025).** The Impact of Pre-adulthood Immersive Media Experiences on Continued Use Intention in Adulthood: An Empirical Study Applying the Value-based Adoption Model and Positive Technology Readiness Index. *Information Society & Media*, 26(1), 143–183.

MANUSCRIPTS UNDER REVIEW

- Kim, G., & Lee, C. (under review).** Socially Visible Sensing in AR Smart Glasses: Outward Recording, Societal Concern, and Wearer Stigma. *Technology in Society*. Manuscript under review.
- Kim, G., Choi, Y., & Lee, C. (under review).** How Much Should Proactive AI Know About Its User? Profile Granularity in AI-Glasses Support for Research Ideation. *CHI 2027*. Manuscript under review.
- Kim, G., Choi, Y., Kim, Y., & Lee, C. (under review).** Goodbyes Matter: Immersive Relational Offboarding for Reducing Loss When AI Companion Interactions End. *CHI 2027*. Manuscript under review.

SELECTED FUNDED RESEARCH PROJECTS

Youth-Focused AI Risk Evaluation Benchmark Development	Researcher · ETRI · 2026
Proactive AI and Human Cognitive Mechanisms in Always-On AI Environments	Researcher · NRF · 2026–present
Personalized AI Story Generation for Children	Researcher · Seoul RISE · 2026–present
Interactive Experiential News Game Research and Development	Researcher · SKKU RISE · 2025–2026
Screen Multimodal Interaction Research · Big Data Analytics Team	Researcher · Samsung Electronics · 2024
Virtual-World Multi-Persona Tendencies and Cognitive Enhancement	Researcher · NRF Korea · 2025
Urban Green Exposure, Mobility, and Health	Researcher · AI Convergence Research Support · 2025
Psychological Counselling Chatbots: Humanities–Engineering Convergence	Researcher · NRF Korea · 2026
News and Social-Media Issue Analysis for the National Health Insurance Service	Researcher · NHIS · 2024

SELECTED RESEARCH AND PROJECTS · PORTFOLIO AND DEMOS

AR Smart Glasses and Socially Visible Sensing	Master’s thesis, 2026
– Used a scenario-based experiment with 369 prospective users to compare outward recording, eye tracking, and AI analysis; wearer stigma was the only resistance factor directly associated with lower intention to use.	
– Third study in a programme of work on AR smart glasses, alongside the hardware-based proactive-AI and AI-companion experiments below.	
See, Sense, Spark — Proactive AI for Creative Research Ideation	SIGGRAPH Asia 2025 Poster
– Built a proactive AI system on XREAL Air 2 Pro smart glasses. The initial study (analytic N = 17) found higher interest-driven curiosity with fine-grained profiles (p = .016, d = 1.30). In the CHI 2027 follow-up (N = 26; 24 complete logs), profile policies sharply changed AI-generated content without detectable differences in participant question breadth or self-reports, motivating the system–participant divergence gap.	
No Time to Say Goodbye — Termination and Emotional Loss in AI Companions	CHI EA 2026
– Built IRO and ran a between-subjects experiment (N = 26; ANCOVA, p = .011), then extended it in Goodbyes Matter (CHI 2027 manuscript under review, N = 56), where advance notice reduced immediate loss without reducing immersion.	
ThinkCap — Critical Reading Interface	Independent Prototype in Development
– Prototyped a browser extension, native macOS overlay, and BLE firmware for line focus, critical marking, annotation, and behavioural logging; physical integration and evaluation remain in development.	
Phodong: AI-Based Phygital Story Creation for Children	HCI Korea 2026 Creative Award Grand Prize
– Developing a SwiftUI/Firebase system that turns child-discovered objects into personalised stories and traces each object from discovery through story inclusion and child response.	
Progressive AR Sound Cues	Research System in Development
– Implemented a DHH-informed Unity/XREAL package with separated event sources, cue policies, head-pose logs, and counterbalanced trials; device validation and participant studies remain pending.	

SELECTED TECHNICAL DEVELOPMENT

3D Gaussian Splatting Reconstruction and Viewer	WebGL / GLSL, 2025
– Built a COLMAP and 3D Gaussian Splatting reconstruction workflow and reimplemented core forward-rasterisation, alpha blending, depth sorting, and camera-aware optimisation in JavaScript and GLSL.	
Neural Models for Visual and Text Classification	PyTorch, 2026
– Designed and evaluated an MLP for Fashion-MNIST and a padding-aware bidirectional LSTM for IMDb sentiment classification, with controlled comparisons of augmentation, regularisation, pooling, and training configurations.	
Interactive Graphics and Geometry Processing	Three.js / WebGL / GLSL, 2025
– Implemented articulated animation and interactive graphics in Three.js, plus half-edge mesh processing, Catmull–Clark subdivision, and derivative-based texture filtering in GLSL.	

AWARDS, EXHIBITIONS, AND CREATIVE WORK

HCI Korea 2026 Creative Award · Grand Prize for Phodong	2026
Selected Team, 2026 Promising Student Startup Team 300+ · Growth Track	2026
GS SHOP Content Leader Scholarship · 13th Cohort	Jan 2025
SKKU Summa Cum Laude · Sungkyunkwan University	Feb 2025
Best Award · 2nd Joint Academic Festival of Convergence Colleges	Aug 2024
Best Presentation Award · Cybercommunication Society Spring Conference	May 2024
Ars Electronica 2024 · Miracle on the Han River, co-creator and exhibitor, Linz	2024
HCI Korea 2024 Creative Award Proceedings · Published contribution	2024
Grand Prize · 2nd Culture and Technology Convergence Major Day Capstone Project Competition	Dec 2023

LEADERSHIP AND ACADEMIC SERVICE

President, 4th Student Council CNT:role, Culture and Technology Convergence	2023
Led student community programmes at Sungkyunkwan University and helped launch the first joint academic festival across five convergence colleges.	

RESEARCH SKILLS

Programming and data	Python; R; Swift; JavaScript; TypeScript; NumPy; pandas
AI, XR, and graphics	PyTorch; scikit-learn; Unity; Unreal Engine; XREAL; SwiftUI; Firebase; WebGL
Research methods	Controlled experiments; preregistration; interaction-log analysis; mixed models; permutation/bootstrap inference; SEM; qualitative coding
Research areas	Human–AI interaction; AI contribution and appropriation; proactive AI; agency and metacognition; AR/XR; trace-based evaluation
Languages	Korean (native); Chinese (proficient — six years of residence in China); English (academic reading, writing, and research communication)

REFERENCES

Prof. Changjun Lee · M.S. advisor · Sungkyunkwan University · changjunlee@skku.edu